

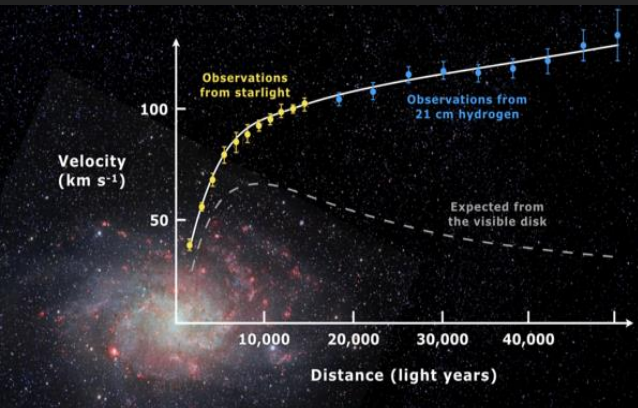


LDMX

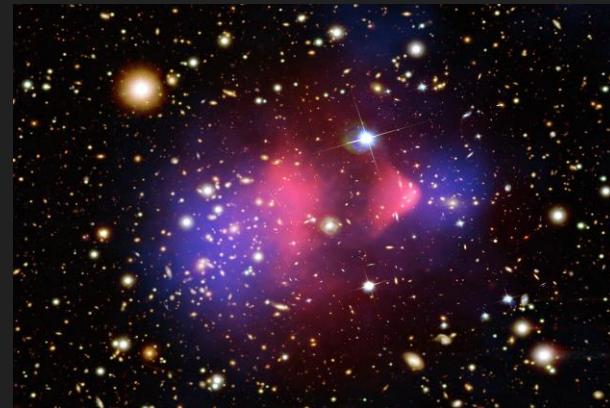
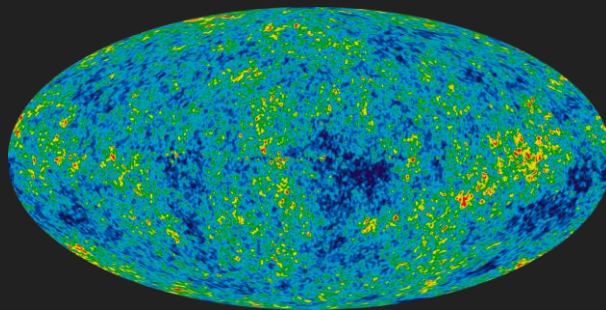
The Light Dark Matter Experiment

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University of Virginia

Dark Matter is out there!

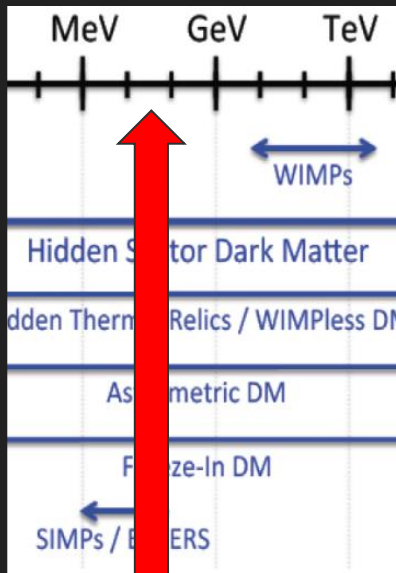


Galaxy Rotation Curves



Bullet Cluster

DM Mass Range

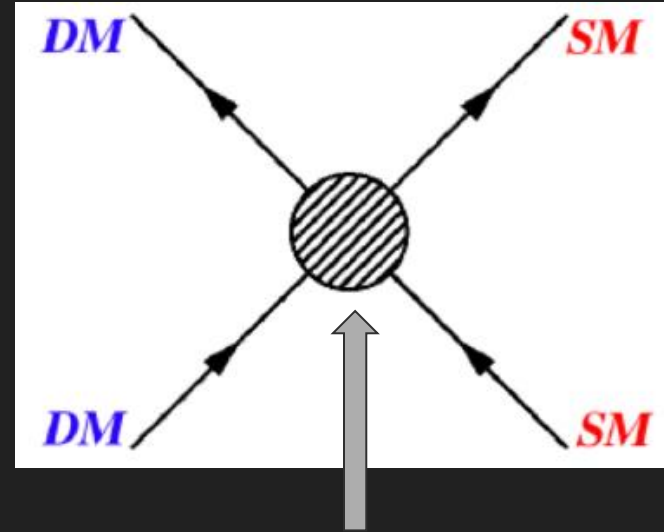


LDMX

Can constrain
with thermal-
freeze out model

Light DM Phenomenology

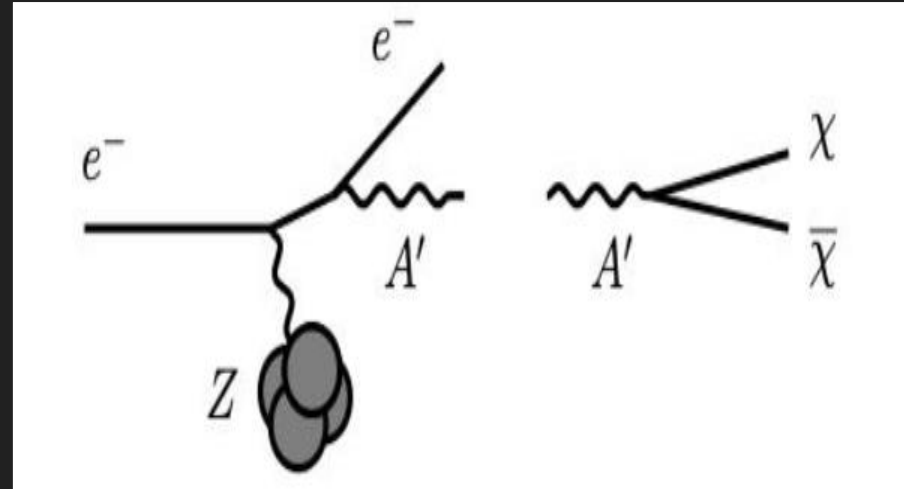
- “Dark Photon” (A') from new broken $U(1)$ symmetry group
- Can theoretically interact with SM particles
- $m_{A'} > 2m_{DM}$ Decays Invisibly!

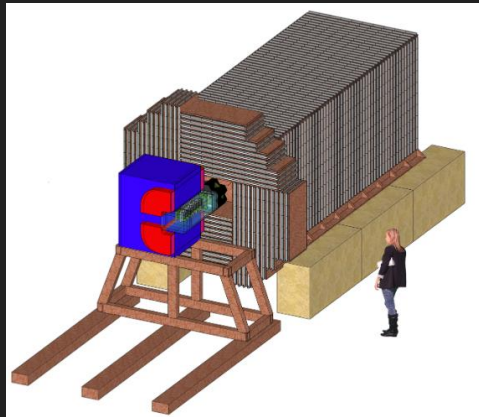
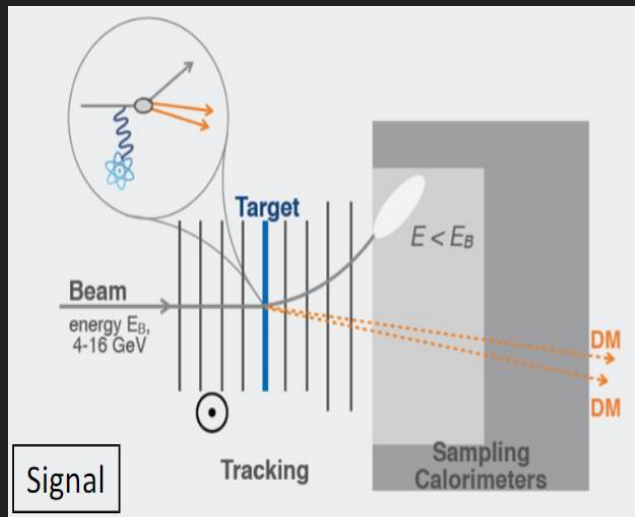


A' would be the link!

Missing Momentum

- A' is heavy! When interacting with SM particle (say an electron), takes most of the energy and then decays *invisibly*
- Dark Bremsstrahlung
- Signal of interaction would be single low energy electron with “missing” energy momentum lost to the A'





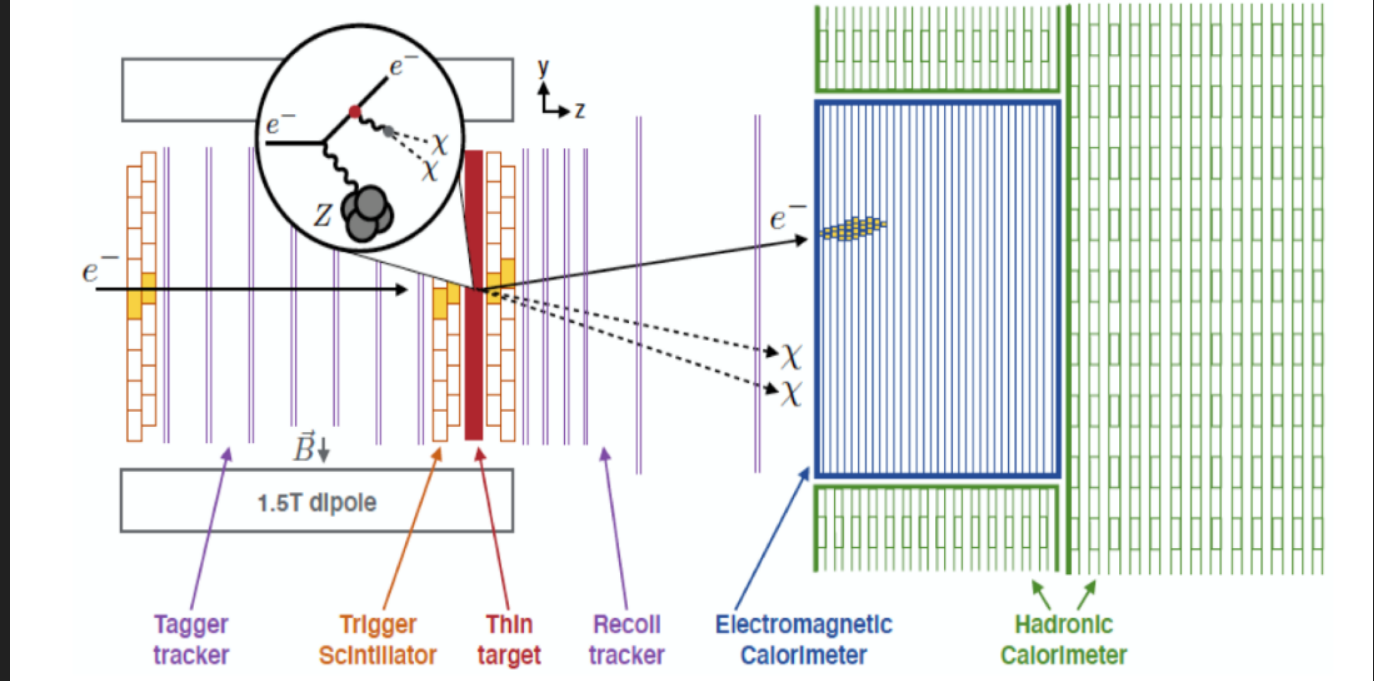
LDMX Concept

- 4-8 GeV electron beam shot at Tungsten target
- Series of detectors and calorimeters look for missing momentum signal, vetoes background events

LDMX Design

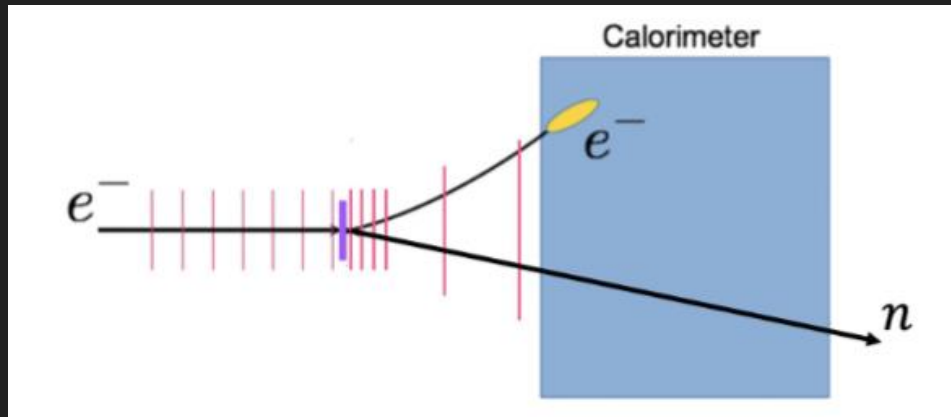
Primary Components

- Tagger
- Trigger
- Recoil Tracker
- ECal
- HCal



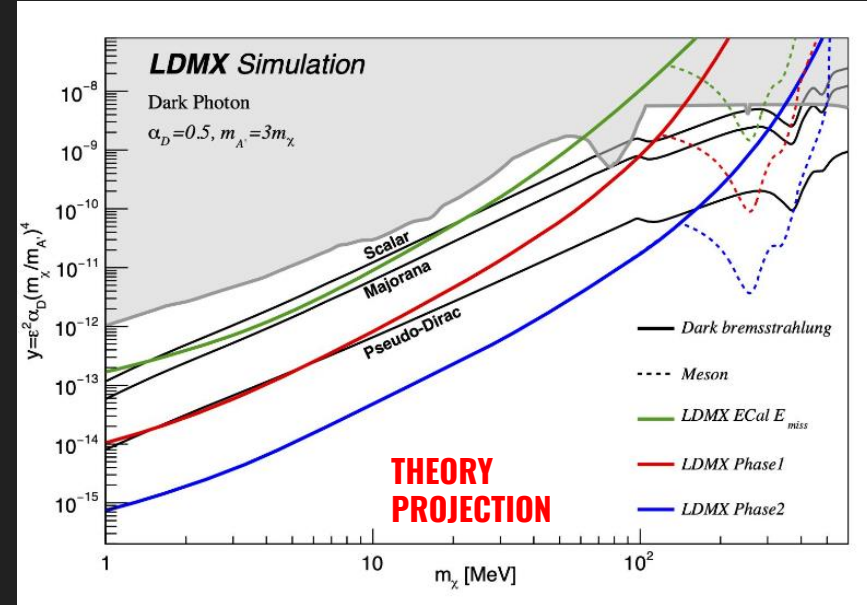
Backgrounds

- Main background, normal bremsstrahlung!
- Vetoing background requires combination of hard cuts and machine learning (Boost Decision Tree)



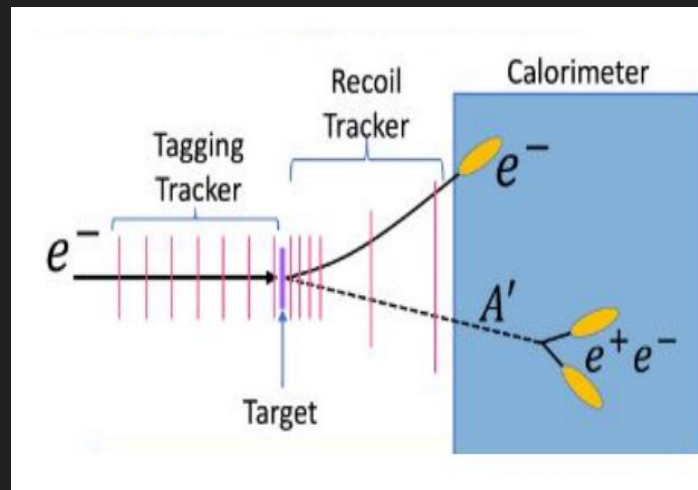
Missing Momentum Reach

- Simulation results:
 - <1 Background Event
 - 30-50% Signal Efficiency
 - 10^{14} Electrons on Target
- **Will improve light dark matter search sensitivity by several orders of magnitude!**



LDMX Visibles Search

- High precision instrumentation in the trackers, Ecal, and Hcal can probe other DM detection modes
- *Visible A' Decay:*
 - $m_A < 2m_{DM}$ Decays Visibly!
 - New signal: low momentum electron + electron and positron
 - Can look for these decays in Ecal and Hcal using Boosted Decision Trees (BDTs)



Visible Regimes

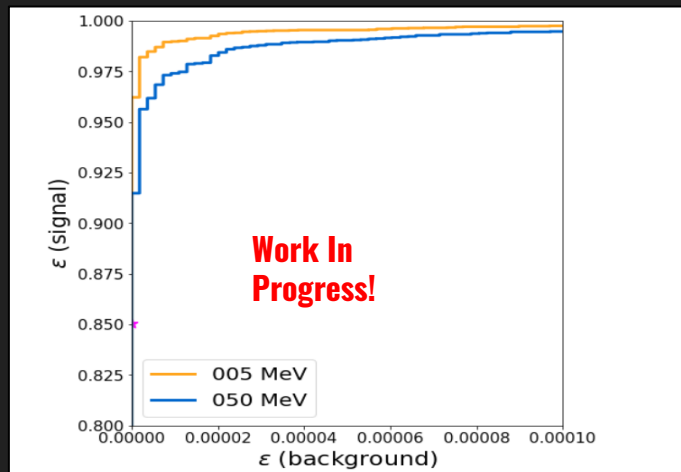
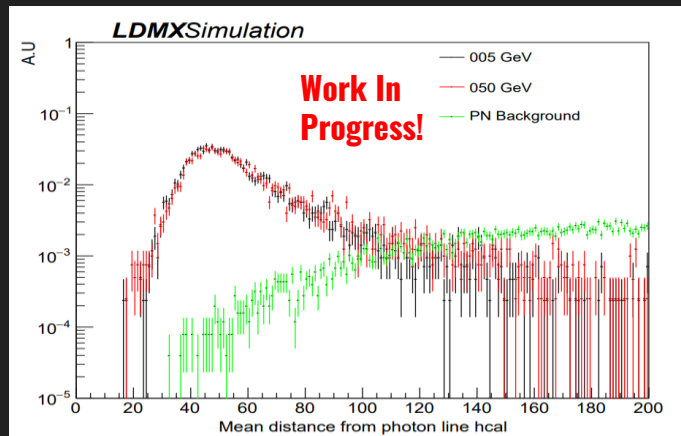
- 3 different BDTs based on where event energy ends up

	ECal _{back half}	HCal	ECal _{back half} + HCal
ECal	>2.5 GeV	<2.5 GeV	—
HCal	<2.5 GeV	>2.5 GeV	—
Intermediate	<2.5 GeV	<2.5 GeV	>2.5 GeV

My focus!

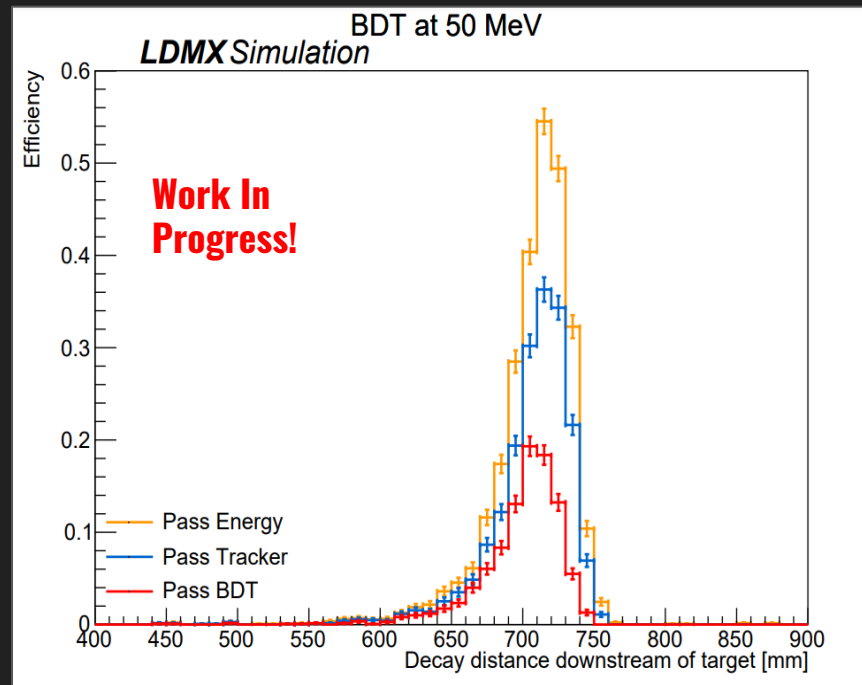
Intermediate BDT

- Hardest region in terms of background, decays in both ECal and HCal
- Combine high performing features from ECal and HCal BDTs (19 total)
- Identifying event features with good signal/background discrimination



Intermediate BDT Simulation Results

- Simulation of $O(10^{14})$ EoT
- **5** passing events at 85% BDT signal efficiency threshold
- When integrated with whole visible veto regime get around **20%** signal efficiency (~slightly less than missing momentum channel)



Conclusion

- LDMX Missing Momentum search can improve light dark matter search sensitivity by several orders of magnitude!
- LDMX Visibles Search offers complementary search channel with potential for low background and competitive signal efficiency
- LDMX is relatively inexpensive, can be completed with **EXISTING** technology, and the required beam infrastructure is under construction at SLAC.



Thank You!



Caltech Fermilab



UNIVERSITY OF MINNESOTA

UCSB
UNIVERSITY OF CALIFORNIA
SANTA BARBARA

SLAC
NATIONAL
ACCELERATOR
LABORATORY



Intermediate BDT Features

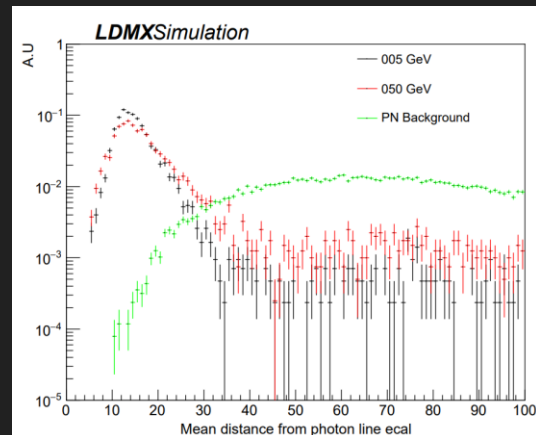
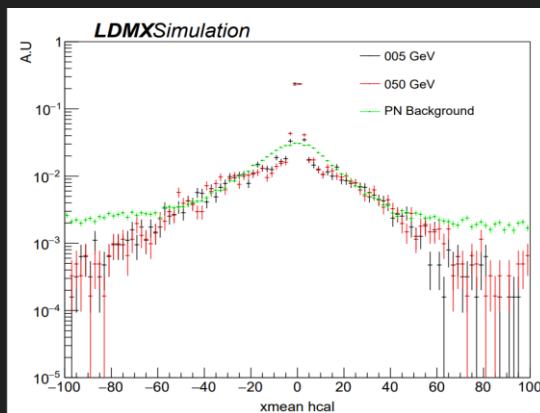
Feature F-Score Rankings

- Mean Photon Distance (ecal)
- Isolated Hits (ecal)
- Mean Photon Distance (hcal)
- Ymean (ecal)
- Downstream Hits (ecal)
- Isolated Energy (hcal)
- Isolated Hits (hcal)
- Zstd (ecal)
- Upstream Energy (ecal)
- Total Energy (ecal)
- Xmean (ecal)
- Isolated Energy (ecal)
- Ystd (ecal)
- Xstd (ecal)
- Xstd (hcal)
- Zstd (hcal)
- Ystd (hcal)
- Ymean (hcal)
- Xmean (hcal)

Best

Worst

BAD separation
between signal
and background!



GOOD separation
between signal
and background!